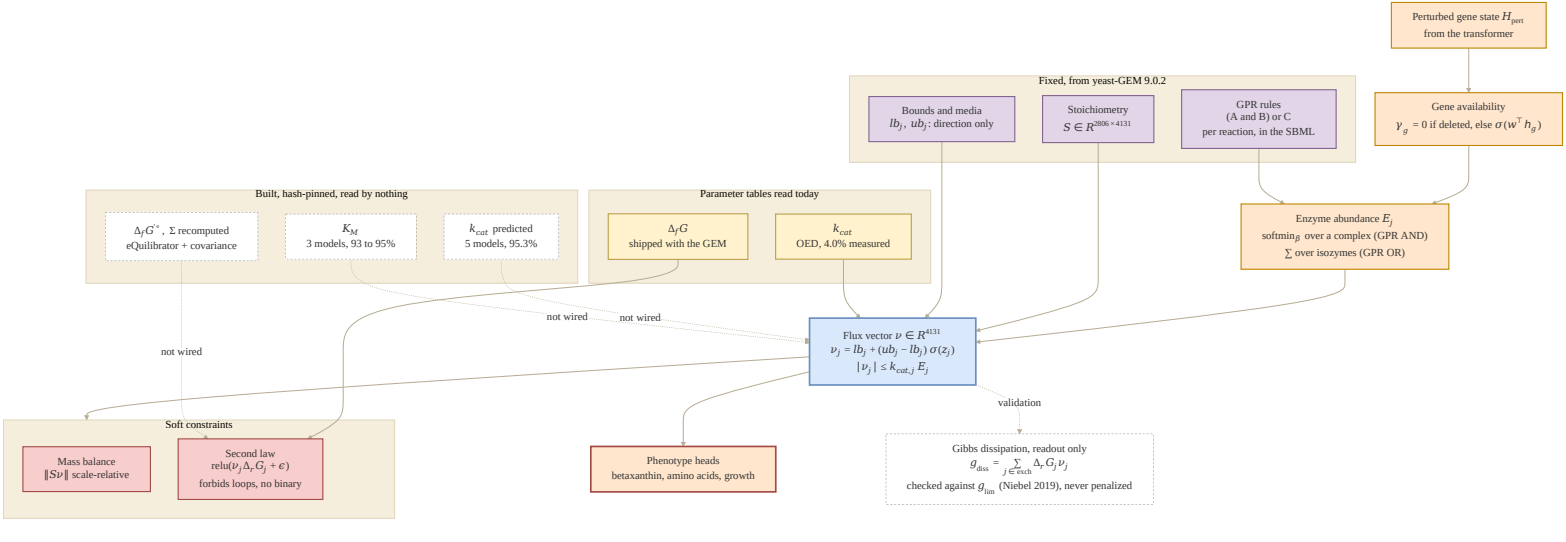

Perturbed gene state $H_{pert}$ from the transformer
Fixed, from yeast-GEM 9.0.2
Bounds and media $lb_j$, $ub_j$: direction only
Stoichiometry $S \in R^{2806 \times 4131}$
GPR rules (A and B) or C per reaction, in the SBML
Gene availability $y_g = 0$ if deleted, else $\sigma(w^\top h_g)$
Built, hash-pinned, read by nothing
$\Delta_r G'^\circ$, $\Sigma$ recomputed eQuilibrator + covariance
$K_M$ 3 models, 93 to 95%
$k_{cat}$ predicted 5 models, 95.3%
Parameter tables read today
$\Delta_r G$ shipped with the GEM
$k_{cat}$ OED, 4.0% measured
Enzyme abundance $E_j$ softmin$_\beta$ over a complex (GPR AND) $\Sigma$ over isozymes (GPR OR)
not wired
not wired
not wired
Flux vector $v \in R^{4131}$ $v_j = lb_j + (ub_j - lb_j)\,\sigma(z_j)$ $|v_j| \le k_{cat,j}\,E_j$
validation
Soft constraints
Mass balance $\|Sv\|$ scale-relative
Second law $\mathrm{relu}(v_j \Delta_r G_j + \epsilon)$ forbids loops, no binary
Phenotype heads betaxanthin, amino acids, growth
Gibbs dissipation, readout only $g_{diss} = \sum_{j \in exch} \Delta_r G_j v_j$ checked against $g_{lim}$ (Niebel 2019), never penalized